INCH-POUND
MIL-PRF-53064C
13 September 2017
SUPERSEDING
MIL-PRF-53064B
6 February 1996

PERFORMANCE SPECIFICATION

HOSE ASSEMBLY, ELASTOMER: LIGHTWEIGHT, COLLAPSIBLE; LIQUID FUEL

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a lightweight, collapsible, elastomer hose assembly for use with liquid fuels (see 6.1).

1.2 Classification. Hose assemblies will be of the following sizes and classes, as specified (see 6.2):

1.2.1 Sizes. The sizes of the hose and hose assemblies, as determined by the nominal inside diameter of the hose, are as follows:

2 inches
3 inches

1.2.2 Classes. The classes of hose and hose assemblies will be as follows:

Class 1 - Coupling Halves, Quick-Disconnect, Cam-Locking Type, Hose Shank; Female Both Ends.
Class 2 - Coupling Halves, Quick-Disconnect, Cam-Locking Type, Hose Shank; Male Both Ends.

Comments, suggestions, or questions on this document should be addressed to U.S. Army Tank-Automotive Research, Development and Engineering Center, ATTN: RDTA-SIE-ES-SI MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or sent by email to mailto: usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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Class 3 - Coupling Halves, Quick-Disconnect, Cam-Locking Type, Hose Shank; Female One End, Male Other End.
Class 4 - Quick-disconnect, sexless couplings

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59326	-	Coupling Halves, Quick-Disconnect, Cam-Locking Type, General Specification for
A-A-59326/10	-	Coupling Half, Cap, Dust, Type IX
A-A-59326/11	-	Coupling Half, Plug, Dust, Type X
A-A-59377	-	Coupling Assembly, Quick-Disconnect, Sexless Type.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35 and JP-8+100 (NATO F-37)

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 173.403	-	Definitions

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(Copies of this document are available online at www.gpoaccess.gov/cfr/index.html.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://www.ansi.org/.)

ASTM INTERNATIONAL

ASTM D380	-	Standard Test Methods for Rubber Hose
ASTM D381	-	Standard Test Method for Gum Content in Fuels by Jet Evaporation
ASTM D413	-	Standard Test Methods for Rubber Property – Adhesion to Flexible Substrate
ASTM D471	-	Standard Test Method for Rubber Property – Effect of Liquids
ASTM D975	-	Standard Specification for Diesel Fuel Oils
ASTM D1149	-	Standard Test Methods for Rubber Deterioration – Cracking in an Ozone Controlled Environment
ASTM D1655	-	Standard Specification for Aviation Turbine Fuels

(Copies of these documents are available from www.astm.org.)

NORTH ATLANTIC TREATY ORGANIZATION (NATO)

AFLP-3747	-	Guide Specifications (Minimum Quality Standards) for Aviation Turbine Fuels (F-24, F-27, F-34, F-35, F-37, F-40 and F-44)

(Copies of this document are available online at http://www.nato.int/.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS-STD-595	-	Colors Used in Government Procurement

(Copies of these documents are available online at http://www.sae.org/.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2). The hose assembly shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Unless otherwise specified herein, materials used shall be selected by the manufacturer. The hose assemblies shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against corrosion and deterioration during storage and operational conditions experienced. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Recovered materials shall be used to the maximum extent practicable (see 4.6.1). Asbestos, cadmium, and radioactive material will not be used in this item. Radioactive material is defined by 49 CFR 173.403, or other radioactive material in which the radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item.

3.3 Design and construction. The hose shall consist of a compounded elastomer inner tube, one or more plies of synthetic fiber reinforcement material, static wire, and compounded elastomer cover (see 4.6.2). The hose shall not leak. The color of the hose shall be sand matte in accordance with SAE AMS-STD-595 color ID 33446. The tube shall be fabricated of an elastomer compounded material which is resistant to hydrocarbon fuels (see 3.3.4).

3.3.1 Static bond. Unless otherwise specified, the hose shall contain a continuous embedded static wire(s) or other electrical conductor(s). The conductor(s) shall provide a continuous electrical bond between the hose ends to include fittings. The maximum resistance shall not be greater than 1.5 ohms per linear foot of hose (see 4.6.13). The bond shall not break when the hose is coiled, sharply bent, or subjected to internal or external pressures as specified.

3.3.2 Reinforcement. The reinforcement of the hose shall be of a carcass of synthetic filament warp yams interwoven with fillers of synthetic filament yams or shall be of one or more plies of woven, braided or spiraled synthetic fiber cord. The reinforcement material shall be free of dirt, knots, lumps, or irregularities of twist. The reinforcement shall be evenly applied and thoroughly impregnated with elastomer so that the plies are bonded to their adjacent layers, tube, and cover, so as to produce a unified hose wall.

3.3.3 Couplings and fittings.

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3.3.3.1 Coupling halves and coupling caps. The coupling halves on Class 1 and Class 2 hoses shall be cam-locking type in accordance with A-A-59326. The female coupling shall be fitted with a dust plug per A-A-59326/11. The male fitting shall be equipped with a dust cap per A-A-59326/10. The plugs and caps shall be attached to the hose end coupling halves by means of chain attachments or other similar attachments (see 4.6.2).

3.3.3.2 Quick-disconnect, sexless couplings. The sexless couplings shall be Class A in accordance with A-A-59377. The couplings shall be reusable.

3.3.4 Fuel and Water compatibility. Finished hoses shall be resistant to, have no deleterious effects upon, and not be degraded by water and the following kerosene-based petroleum fuels. Meeting the hose physical and chemical properties requirements with the specified test fluid shall provide sufficient evidence that the finished hoses conform to the fuel compatibility requirement.

a. MIL-DTL-83133 (JP-8) (NATO F-34).

b. MIL-DTL-5624 (JP-5) (NATO F-44).

c. ASTM D975 (Diesel-US commercial).

d. MIL-DTL-83133 (NATO F-35).

e. ASTM D1655 (Jet A-1).

f. AFLP-3747 (NATO F-24).

3.4 Performance.

3.4.1 Diameter weight and length. The length of hose assembly shall be as specified (see 6.2) with a tolerance of ±0.5 inches. Hose inner diameter (ID) and weight shall be as specified in table I (see 4.6.2).

TABLE I. Diameters and weights.

Hose ID	Weight per linear foot without couplings
2 inches	0.60 pounds (max)
3 inches	0.95 pounds (max)

3.4.2 Hydrostatic pressure. The hose and hose assembly shall be tested in conformance to ASTM D 380 per elongation, twist, and proof pressure, using water as the test fluid (see 4.6.3). The hose and hose assembly shall be tested at the proof pressure listed in table II in pounds per square inch (psi). While under pressure, the hose assembly shall not leak, and there shall be no indication of imperfections, slippage or pull-out of the hose from the coupling. The length of the hose shall not change more than ±2 percent.

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TABLE II. Pressure for hose assembly.

Size ID	Working Pressure (minimum)	Proof pressure (minimum)	Burst pressure (minimum)
2 inches	75 psi	150 psi	300 psi
3 inches	75 psi	150 psi	300 psi

3.4.3 Minimum burst. The hose and hose assembly shall be tested for burst pressure. When tested, each hose size shall have a minimum burst pressure as specified in table II (see 4.6.4).

3.4.4 Tensile strength. The tensile strength of the tube and cover shall not be less than 1250 psi and 1500 psi respectively (see 4.6.5).

3.4.5 Tensile strength after immersion. The tensile strength of the tube and cover after immersion shall not be less than 600 psi (see 4.6.6).

3.4.6 Ultimate elongation. The ultimate elongation of the tube and cover shall be not less than 300 percent, and the elongation after immersion shall be not less than 200 percent (see 4.6.7).

3.4.7 Original adhesion and adhesion after immersion. The adhesion between the tube and plies, between the plies, and between the cover and plies before and after immersion shall not be less than 10 pounds force (lbf) (see 4.6.8) and 6 lbf (26.7 N) respectively (see 4.6.9).

3.4.8 Low temperature flexibility. Low temperature flexibility shall be determined at a minimum pressure of 200 psi, there shall be no evidence of cracking, or lack of flexibility, and the hose shall not leak (see 4.6.10).

3.4.9 Ozone resistance. The cover material shall be viewed under 7-power magnification, the cover shall show no evidence of cracking, checking, or other irregularities at the end of the exposure time (see 4.6.11).

3.4.10 Existent and heptane-washed gum content. The hose shall be tested for gum content, the existent gum content shall not be more than 0.0175 pounds per gallon (lb/gal) of test fluid. The heptane-washed gum content shall not be more than 0.00042 lb/gal (see 4.6.12).

3.4.11 Electrical resistance. The electrical resistance of the hose shall be determined from coupling to coupling, and shall not exceed 1.5 ohms per linear foot of hose (see 4.6.13).

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3.5 Hose marking. Hose and hose assemblies shall be permanently marked at 10 foot intervals or by continuous impression branding, provided that the pattern shall be repeated every 5 feet or less. The letters shall be a minimum of 0.18 inches high. Marking color shall contrast with the hose cover color. Each hose shall be marked with a longitudinal stripe of a contrasting color, not less than 0.125 inches wide, and extending the full length of the hoseline. The marking may be within the longitudinal stripe, provided the stripe is at least 0.2 inches greater than the height of the letters. The marking may substitute for the longitudinal stripe, provided it is continuous and extends the full length of the hoseline. Labeling and longitudinal stripe shall remain clear, distinct and legible for the life of the hoseline. Handling, bending, water, oil, and environmental effects shall not adversely affect the marking or stripe. The labeling shall contain the following information unless otherwise specified (see 6.2):

a. Date of manufacture (quarter and year).

b. Manufacturer's name or trademark.

c. "MIL-PRF-53064".

d. "LIQUID FUEL".

3.6 Workmanship. The hose assembly shall conform to the quality requirements as specified herein. There shall be no cuts, nicks, bruises, blisters, lumps, or sharp edges on the hose assembly. The hose assembly shall be clean and smooth. The hose assembly shall have no other defects affecting appearance or serviceability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Quality conformance inspection (see 4.3).

4.2 First article inspection. The first article shall be examined as specified in 4.4, in the order shown. The first article shall be subjected to the tests of 4.6.3 through 4.6.12 (see table III).

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TABLE III. Hose assembly tests.

Test	Requirement	Validation	First article	Quality conformance	Individual
Hydrostatic pressure	3.4.2	4.6.3	X	X	X
Minimum burst	3.4.3	4.6.4	X	X	
Tensile strength	3.4.4	4.6.5	X	X	
Tensile strength after immersion	3.4.5	4.6.6	X	X	
Ultimate elongation	3.4.6	4.6.7	X	X	
Ultimate elongation after immersion	3.4.6	4.6.7	X	X	
Original adhesion	3.4.7	4.6.8	X		
Adhesion after immersion	3.4.7	4.6.9	X	X	
Low temperature flexibility	3.4.8	4.6.10	X	X[1]	
Ozone resistance	3.4.9	4.6.11	X		X
Existent and heptane washed gum content	3.4.10	4.6.12	X		
Electrical resistance	3.4.11	4.6.13		X	

Notes:
1. One lot testing required for each 10 lots submitted.

4.3 Conformance inspection. Conformance inspection shall consist of the examinations of 4.4 and the tests of 4.5.

4.4 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQ Z1.4. Each sample selected shall be inspected as specified herein for the defects listed in table IV and for the tests marked with an “X” in table III. Test sections cut from the hose shall be of sufficient length for the performance of their required test.

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TABLE IV. Examinations.

Category	Defect	Method of examination
Major:		
101	Material not as specified (see 3.2)	Visual
102	Materials not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment (see 3.2).	
103	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the hose assembly (see 3.2).	Visual
104	Hose not as specified (see 3.3).	Visual
105	Color of hose not as specified (see 3.3).	Visual
106	Tube not as specified (see 3.3).	Visual
107	Hose cover scuffed, or bruised, so that braid is exposed (see 3.3).	Visual
108	Hose cover not clean and smooth (see 3.3).	Visual
110	Fittings and couplings not as specified (see 3.3.3.1 and 3.3.3.2).	Visual
111	Dimensions not as specified (see 3.2).	Visual
112	Workmanship not as specified (see 3.6).	Visual
113	Safety marking not as specified (see 3.5).	Visual
114	Electrical resistance not as specified (see 3.4.11).	SIE[1]

Notes:
1. SIE = Standard inspection equipment.

4.5 Individual tests. Each hose assembly produced shall be subjected to those tests marked with an "X" in the Individual column of table III.

4.6 Method of inspection.

4.6.1 Materials. Conformance to 3.2 and 3.3.1 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.3, 3.3.3.1, 3.3.3.2, 3.4.1, 3.5, and 3.6 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with SIE.

4.6.3 Hydrostatic pressure. To determine conformance to 3.4.2, the hose assembly shall be tested to a proof pressure of 150 psi. The electric bond shall be determined while pull test pressure is applied. Elongation shall be ±2 percent. Twist shall be 20° per foot for 2 inch hose and 8° per foot for 3 inch hose.

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4.6.4 Minimum burst. To determine conformance to 3.4.3. The hose and hose assembly shall be tested in conformance to ASTM D380 for burst pressure. The minimum burst pressure shall be 300 psi.

4.6.5 Tensile strength. To determine conformance to 3.4.4, the tensile strength for the tube and cover shall be determined in conformance to ASTM D471, and shall not be less than 1250 psi and 1500 psi respectively.

4.6.6 Tensile strength after immersion. To determine conformance to 3.4.5, the tube and cover shall be tested in conformance to ASTM D471 at a temperature of 73±2 °F, at an immersion period of 70 hours and the immersion fluid shall be reference fuel D. After immersion, the tensile strength shall not be less than 600 psi.

4.6.7 Ultimate elongation. To determine conformance to 3.4.6, the ultimate elongation of the tube and cover before and after immersion shall not be less than 300 percent before, and 200 percent after immersion.

4.6.8 Original adhesion. To determine conformance to 3.4.7, the original adhesion shall be tested in conformance with the machine method of ASTM D413, and shall not be less than 10 lbf.

4.6.9 Adhesion after immersion. To determine conformance to 3.4.7, the hose assembly shall be tested in conformance to ASTM D413, for 70 hours at 93±2 °F in reference fuel C. Within 15 min of removal from the test fluid, the adhesion shall not be less than 6 lbf.

4.6.10 Low temperature flexibility. To determine conformance to 3.4.8, the hose shall be tested in accordance with ASTM D380, at a minimum pressure of 200 psi, at a temperature of -25±2 °F.

4.6.11 Ozone resistance. To determine conformance to 3.4.9 the cover shall be prepared in accordance with ASTM D1149, procedure B1 and tested in accordance with ASTM D1149. The cover shall be mounted in a 20 % elongated position, conditioned for 29 hours in an ozone-free atmosphere. After conditioning, the cover shall be exposed for 72 h at 104±1 °F at an atmosphere containing 50±10 parts per hundred million (pphm) of ozone.

4.6.12 Existent and heptane-washed gum content. To determine conformance to 3.4.10, the hose shall be tested in accordance with ASTM D381. A hose not less than 14 in. long shall be stoppered at one end with a 2 in. plug and secured with a clamp. The hose shall be filled with Reference Fuel D of ASTM D471 and similarly secured at the opposite end. A sample of the neat fuel shall be retained for later use. The hose shall be tested for a period of seven days by agitating the fuel by means of movement every 24 hour period. The agitating process is performed by moving the hose from a vertical to a horizontal position at a rate of 2 cycles per minute for 5 minutes at each 24 hour period. At the end of the seven day storage period, the fuel shall be agitated in the same manner as above for 5 minutes and then immediately removed from the hose.

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Two aliquots of fuel from the hose shall be tested. The existent/unwashed gum content of the fuel shall be determined in accordance with ASTM D381 (air/steam jet vaporizing medium) and an evaporation time of 45 minutes. The residue obtained from the existent/unwashed gum determination, shall then be used to determine the heptane-washed gum content in accordance with ASTM D381. The retained neat fuel sample shall be tested with the same methods. The existent/unwashed gum content from the neat fuel shall be subtracted from the existent/unwashed gum content from the fuel used in the soak test. The heptane-washed gum content from the neat fuel shall be subtracted from the heptane-washed gum content from the fuel used in the soak test.

4.6.13 Electrical resistance. To determine conformance to 3.4.11, the electrical resistance of each hose shall be determined from coupling to coupling with an ohmmeter, and shall not exceed 1.5 ohms per linear foot of hose.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. This lightweight collapsible hose is intended for use with forward area refueling systems which discharge liquid fuels such as aviation fuel, jet fuel, and diesel fuel.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Size of hose assembly required (see 1.2.1).

c. Class of coupling halves (see 1.2.2).

d. When a first article is not required for inspection (see 3.1).

e. Length of hose required (see 3.4.1).

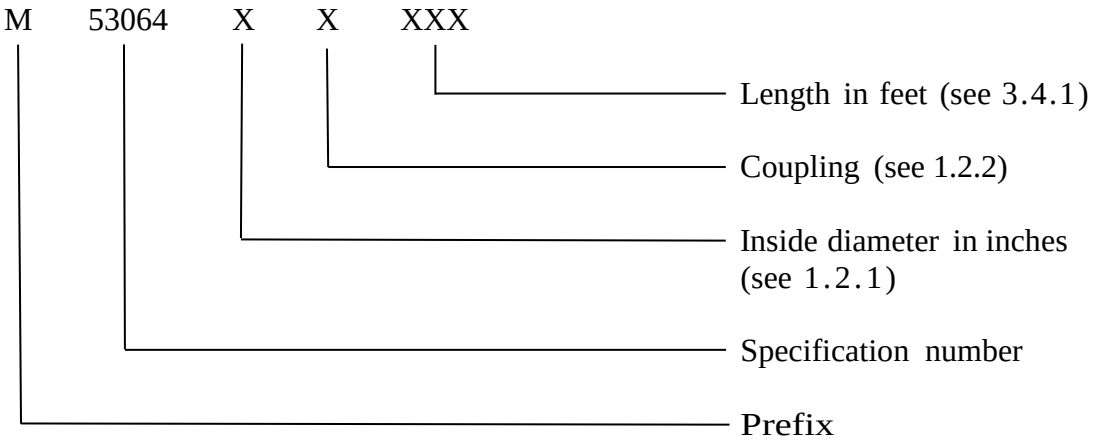
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f. Hose marking (see 3.5).

g. Selection of applicable level of packaging (see 5.1).

h. PIN number (see 6.3).

6.3 Part or identification system. The part or identification number (PIN) of hose assemblies covered by this specification will be designed in the following form (see 6.2).



6.4 Subject term (key word) listing.

Couplings
Cover
Gasoline
Tube

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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